

Faculty of Mathematics and Computer
Department of Computer Science

Bachelor's Project Report

Project Title:
Design of an Intelligent Industrial Equipment Health Monitoring System Based on Deep Learning and Fuzzy Inference System

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Abstract

Industrial equipment failure, particularly in bearings, is a major cause of unexpected production line downtime, increased maintenance costs, and reduced productivity in the mining industry. Timely fault detection and equipment health assessment play a crucial role in properly planning maintenance activities and preventing the escalation of failures. However, many existing systems are limited to fault classification and do not provide sufficient information to support maintenance experts' decision-making.

In this project, an intelligent hybrid system based on deep learning and a fuzzy inference system has been designed and implemented. In the first phase, using bearing vibration data (the standard CWRU dataset with a 48 kHz sampling rate), the fault type is identified by a three-block Convolutional Neural Network (CNN). The final model achieved an accuracy of 97.07% (with a peak of 97.17%). Then, the model's Softmax output along with fault severity weights is fed into a Mamdani Fuzzy Inference System with 44 comprehensive rules to calculate the Health Index, Fault Severity, and Risk Level.

The analysis results are displayed in an interactive bilingual (Persian/English) Streamlit dashboard, allowing experts to view equipment status in a simple, interpretable, and prioritized manner. The system architecture is designed to be extensible to other industrial equipment.

1. Introduction

1.1. Problem Statement

Industrial equipment is one of the most critical assets of production lines in the mining industry. Sudden equipment failure, especially in rotating components like bearings, can lead to prolonged downtime, significant financial losses, and even safety hazards. In many industrial units, equipment condition assessment is still based on periodic inspections, manual analysis of vibration signals, and expert experience. Although useful, these methods face limitations such as delayed diagnosis, dependency on individual expertise, and a lack of quantitative and interpretable indicators for decision-making.

In recent years, the use of deep learning-based methods in vibration data analysis has significantly increased fault diagnosis accuracy. However, most deep learning models only generate the fault class label, and their outputs are not directly interpretable or usable by maintenance experts in the repair decision-making process. This gap between "accurate

diagnosis" and "practical decision support" has been the primary motivation for designing the present hybrid system.

1.2. Importance of the Subject

In the mining industry, such as the National Iranian Copper Industries Company (Sarcheshmeh Copper Complex), equipment reliability plays a decisive role in production process continuity and maintenance cost control. The present project aims to develop an intelligent equipment health monitoring and maintenance decision support system, striving to provide experts with interpretable indicators such as Health Index, Fault Severity, and Risk Level. This approach can provide the basis for improving maintenance planning, reducing unplanned downtime, and increasing equipment reliability.

1.3. Project Objectives

The main objective is to design and implement a hybrid system that combines a CNN for bearing fault diagnosis and a Fuzzy Inference System for converting outputs into interpretable indicators. Specific objectives include:

Design and train a lightweight CNN model for classifying 10 bearing condition classes based on CWRU data.

Develop a Mamdani Fuzzy Inference System to calculate Health Index, Risk Level, and Urgency.

Implement a real-time prediction function and an interactive Streamlit dashboard.

Provide an extensible architecture that can be generalized to other equipment.

2. Proposed Methodology

2.1. Overall System Architecture

The proposed system consists of four main components: (1) Vibration data acquisition and preprocessing, (2) Deep learning engine based on CNN for fault type identification, (3) Fuzzy inference system for calculating health and risk indicators, and (4) Intelligent monitoring dashboard and decision support. The data flow is: Vibration signal → Preprocessing and conversion to 32×32 image → CNN (Fault type + Softmax) → Health Index calculation → Fuzzy System (Risk Level / Urgency) → Dashboard display.

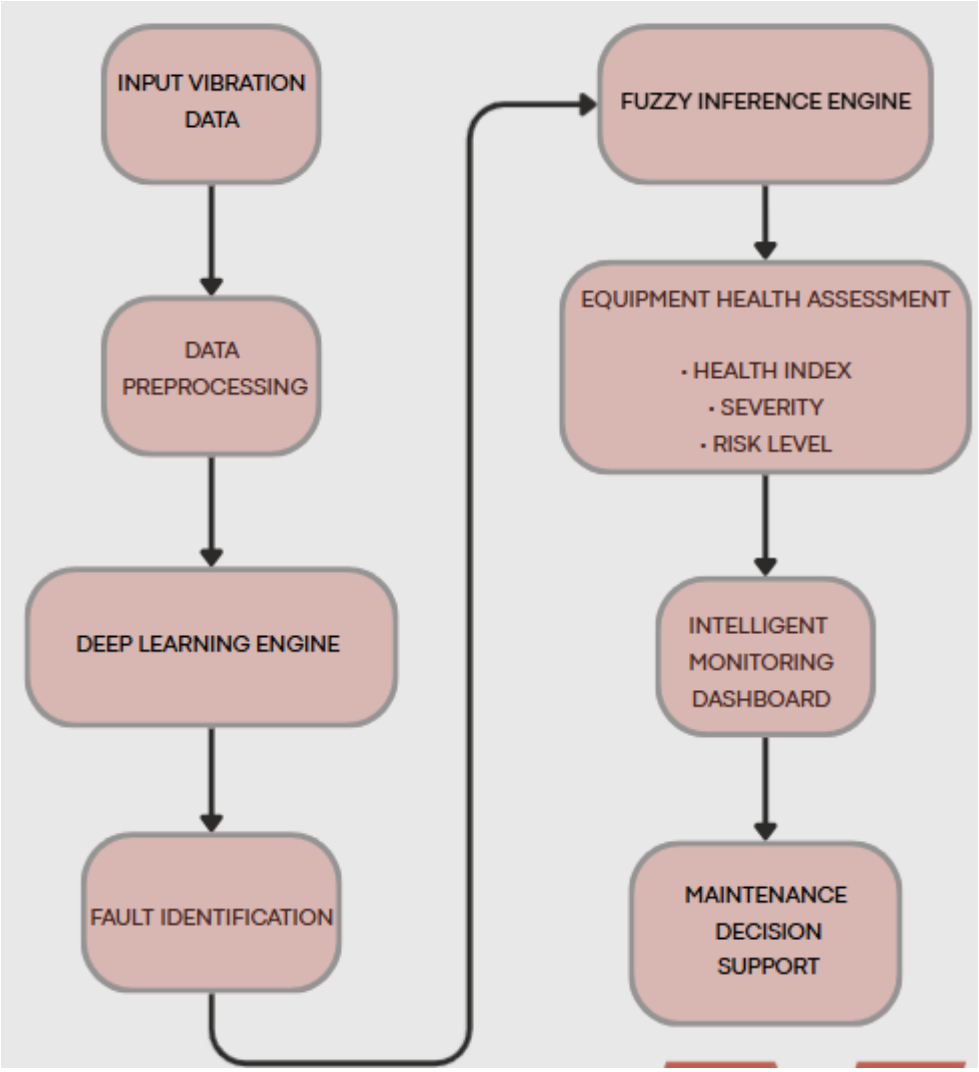


Figure 1: Overall Architecture of the Proposed System

2.2. Dataset and Preprocessing

The standard Case Western Reserve University (CWRU) dataset with a 48 kHz sampling rate was used. It includes 10 bearing condition classes:

Normal (Healthy)

Ball_007, Ball_014, Ball_021 (Ball faults)

IR_007, IR_014, IR_021 (Inner Race faults)

OR_007, OR_014, OR_021 (Outer Race faults)

Each signal window was normalized using the Z-score method and converted to a 32×32 image. The data was split using stratified sampling with an 80% training and 20% testing ratio (approx. 4600 total samples).

2.3. Convolutional Neural Network (CNN) Architecture

The CNN model consists of three consecutive convolutional blocks and a classifier. Each block includes:

2D Convolutional layer (3×3 filter, padding=1)

Batch Normalization

ReLU activation function

MaxPooling (2×2)

Dropout (0.25)

The number of filters in the three blocks is 32, 64, and 128, respectively. The classifier includes three fully connected layers with dimensions of 256, 128, and the number of classes (10). BatchNorm, ReLU, and Dropout (0.5) are used in the intermediate layers. The model was trained using the Adam optimizer and CrossEntropyLoss. The initial learning rate was adjusted using ReduceLROnPlateau. After training, the model achieved a final accuracy of 97.07% (peak 97.17%) on the test set.

2.4. Fuzzy Inference System

The Softmax output along with fault severity weights is used to calculate the Health Index:

$$:\mathrm{HI} = \sum\left(probability_i \times severity_weight_i\right)$$

Then, four input variables are fed into the Mamdani Fuzzy Inference System (**44 comprehensive rules**): Health_Index, Confidence, Fault_Severity, and HI_Trend. The system generates two outputs: Risk_Level and Urgency, which determine the overall state and recommended maintenance action.

2.5. General Algorithm Pseudocode

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Algorithm: Intelligent Bearing Fault Diagnosis Pipeline
Input:  Vibration signal window (32×32), HI_history (optional)
Output: Fault_Type, Health_Index, Risk_Level, State, Recommended_Action

1.  Normalize signal_window (zero-mean, unit-variance)
2.  Convert to tensor and feed to CNN model (BearingCNN)
3.  Compute Softmax probabilities → predicted_class
4.  HI ← Σ (probability_i × severity_weight_i)
5.  Confidence ← max(probability) × 100
6.  Compute HI_Trend from history buffer (if available)
7.  Fault_Severity ← severity_weight[predicted_class]
8.  Feed (HI, Confidence, Fault_Severity, HI_Trend) to Mamdani FIS
9.  Defuzzify → Risk_Level, Urgency
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10. Map Risk_Level & Urgency to State and Recommended_Action
11. Return results to dashboard / real-time API
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2.6. Dashboard and Real-time Prediction

The predict_bearing_fault_real_time function allows receiving a new signal window and returning complete results. The Streamlit dashboard features language switching (Persian/English) and allows monitoring multiple bearings, displaying color-coded risk cards, and viewing equipment details.

3. Results

3.1. CNN Model Performance

The final model achieved an accuracy of 97.07% on the test set. The best accuracy observed during training was 97.17%. The confusion matrix and training/test curves are presented below.

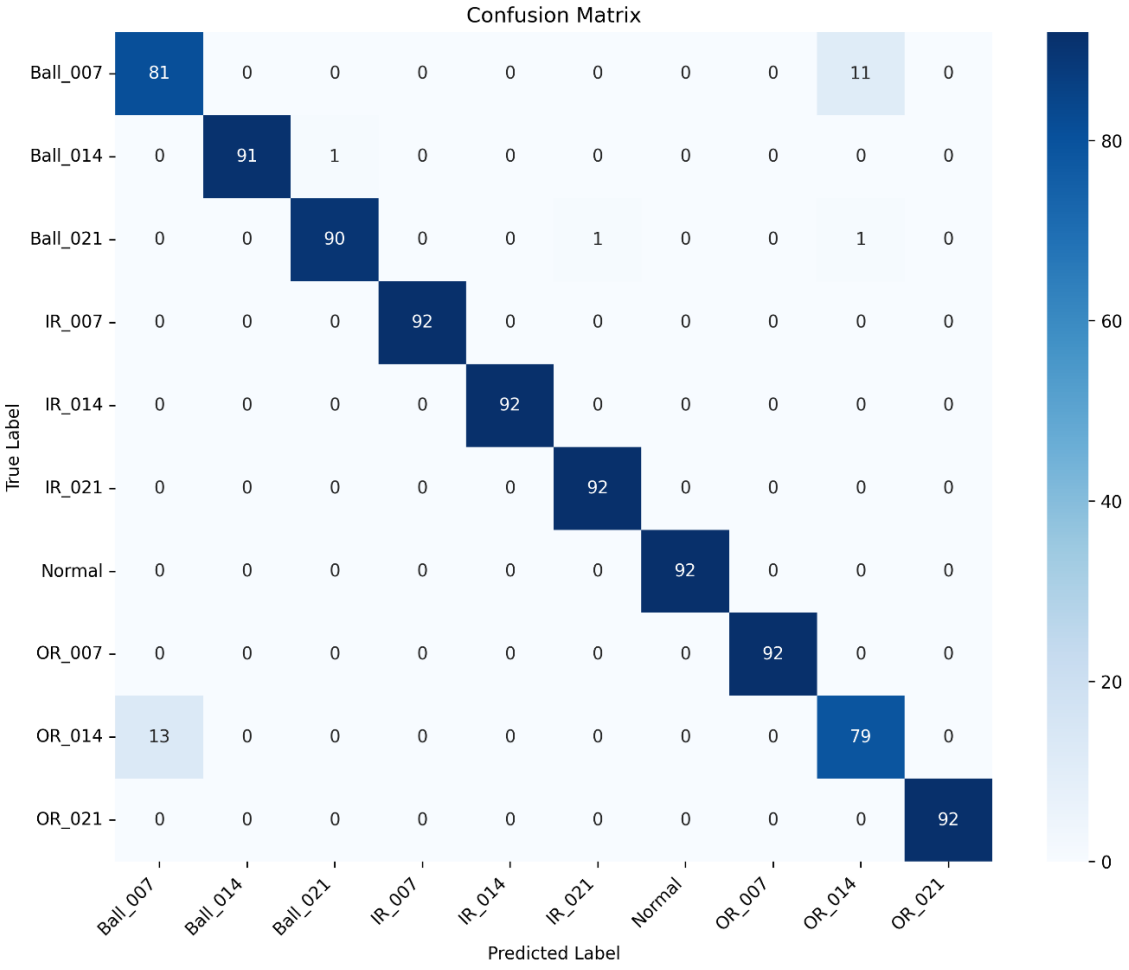


Figure 2: Confusion Matrix

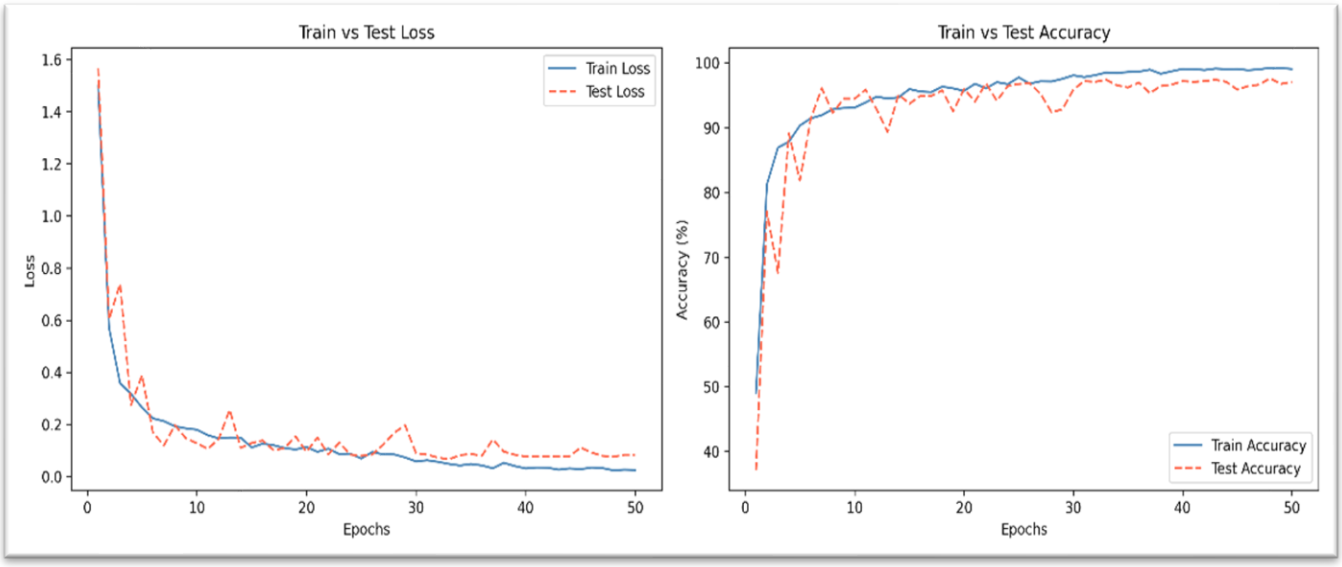


Figure 3: Train/Test Curves

3.2. Intelligent Monitoring Dashboard

The designed dashboard, built with Streamlit, provides an interactive and bilingual (Persian/English) user interface. It allows users to select test data samples, view the frequency spectrum matrix, class probability distributions, and receive fuzzy indicators (Risk and Urgency) with maintenance recommendations.

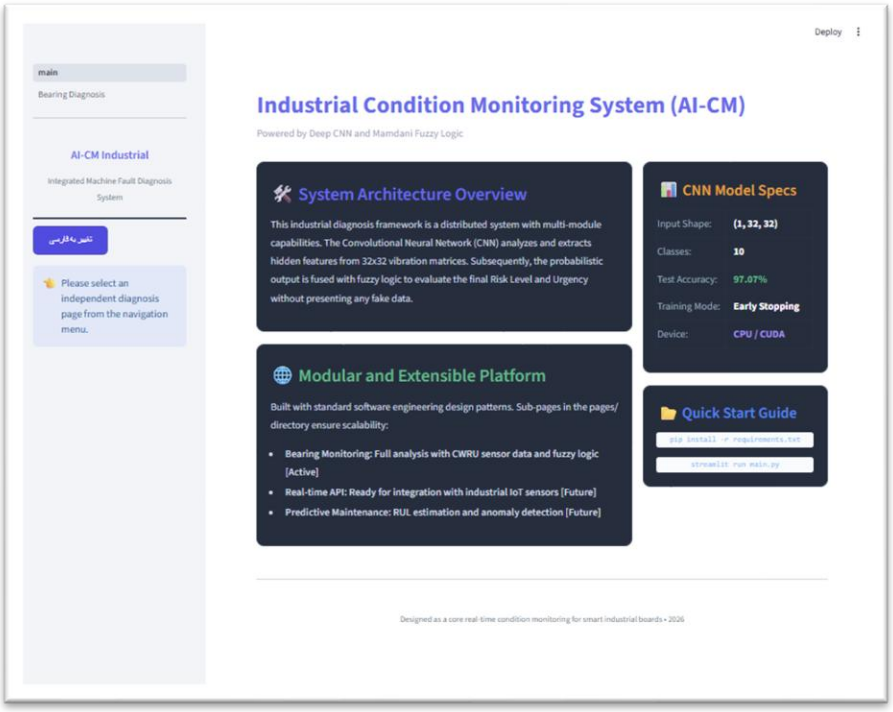


Figure 4: Dashboard Main Page

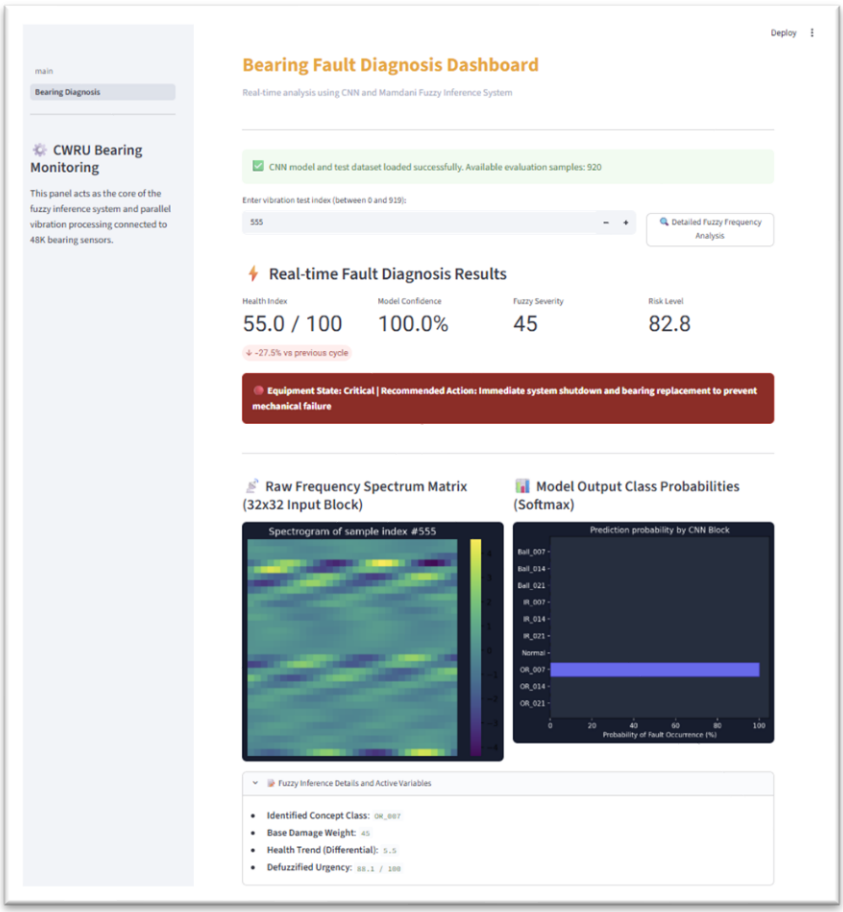


Figure 5: Dashboard Analysis Page

3.3. Summary of Results

The results demonstrate that combining CNN and a fuzzy system can accurately diagnose fault types and generate useful operational indicators for maintenance experts. The high accuracy of the CNN (over 97%) and interpretability of fuzzy outputs make this system a suitable tool for industrial equipment health monitoring.

4. Conclusion and Future Work

In this project, an intelligent hybrid system for industrial bearing health monitoring was designed and implemented. The CNN model achieved an accuracy of over 97%, and the fuzzy inference system converted its outputs into interpretable indicators (Health Index, Risk Level, Urgency). The interactive and bilingual dashboard facilitated the practical application of the results.

Future work recommendations include:

- Collecting and utilizing real vibration data from the Sarcheshmeh Copper Complex.
- Extending the model to other equipment (motors, gearboxes, pumps, fans).
- Adding Remaining Useful Life (RUL) prediction capability.
- Integration with existing factory condition monitoring systems.
- Pilot deployment in a production line.
